

ERC Synergy Grant 2022

Research proposal[Part B2]

Neurological MechanismS of Injury, and Sleep-like cellular dynamics (NEMESIS)

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Section a. State-of-the-art and objectives

Focal brain disorders, including stroke, trauma, and epilepsy, are the main causes of disability and loss of productivity in the world, and carry a cumulative cost in Europe of about 500 billion euro/year. Physicians still largely diagnose and treat these conditions as if they were caused by local dysfunction due to the pathological process. However, there is growing evidence that, in most focal brain disorders, clinical symptoms reflect widespread network-related abnormalities that could be ameliorated by normalizing patterns of activity through “circuit-based” stimulation. This form of therapy is currently limited by numerous factors: lack of knowledge about the underlying circuitries and their relevance to symptoms; a strategy for mapping circuit abnormalities in single patients; and, most importantly, a principled understanding of where and how to stimulate the brain.

NEMESIS from the Ancient Greek “Νέμεσις” is the goddess who enacts retribution against those who succumb to hubris. In Greek the word translates in “give what is due”. NEMESIS aims to “give to an injured brain what is missing”, i.e., **restore through stimulation normal activity in dysfunctional brain circuitries**. By

synergizing four outstanding research groups, concepts, and technologies, NEMESIS will first characterize the effects of focal injury at multiple spatial and temporal scales from local circuits to whole brain. Next, through the combination of observational (fMRI, EEG, calcium imaging, electrophysiology) and causal methods (electro-magnetic stimulation, optogenetics) NEMESIS will test the hypothesis that disconnected networks lie in a “sleep-like state” characterized by the intrusion of slow waves that impair communication. Thirdly, NEMESIS will create whole-brain models of structure/function to predict the effect of individual lesions and simulate novel stimulation protocols aimed at “re-awakening” the disconnected brain. Finally, guided by animal experimental findings and computational modelling, NEMESIS will develop proof-of-concept treatments that combine circuit-based stimulation and rehabilitation to restore normal activity and behaviour, alas “give back what is due” (**Fig.1**). This grant was recommended for funding in 2020, but eventually it was not (2nd on the waiting list). This re-submission is significantly strengthened by a wealth of new preliminary results that make our central hypothesis stronger and the whole proposal more feasible.

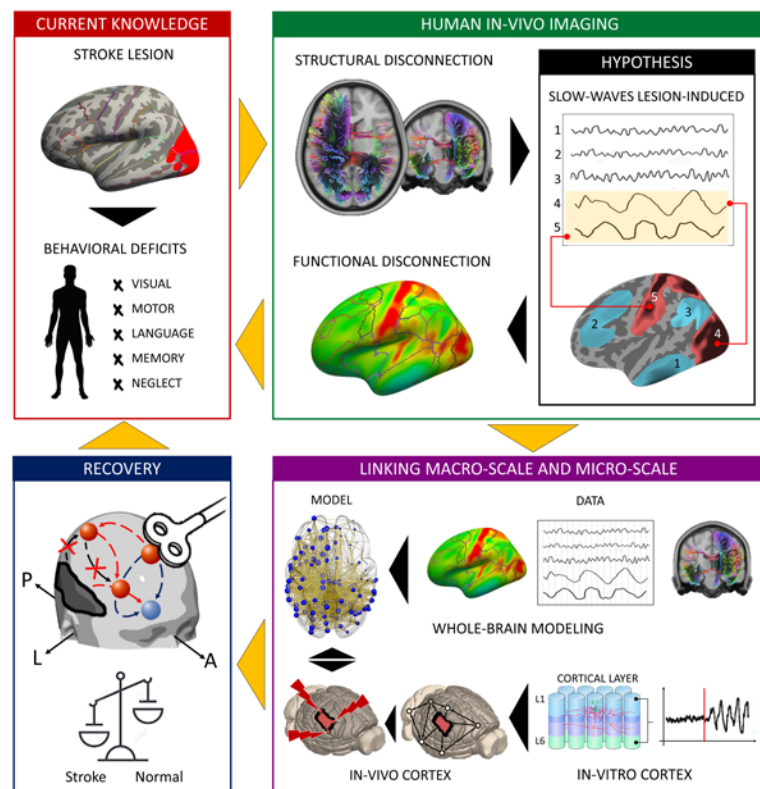


Fig. 1. NEMESIS' vision. Focal lesions cause different combinations of behavioural deficits. The prediction and recovery of these deficits is incomplete. Recent studies from our team show widespread patterns of structural and functional disconnection that empirically predict impairment and recovery of function. Our hypothesis is that these macroscale patterns underlie a specific mechanism: sleep-like slow-wave activity that develops near the lesion and spreads through the network impairing processing. Using individualized measures of functional-structural disconnection and slow waves we will build whole-brain computer models of the lesion and network activity to solve the problem of ‘where’ and ‘how’ to stimulate. Slice and in-vivo animal experiments will replicate network abnormalities and find protocols to normalize these abnormalities. Animal and computer models will inform proof-of-concept treatment protocols that combine stimulation with rehabilitation to normalize activity and behaviour in individual patients, or “give back to the brain what is due” (“Νέμεσις”).

Why is this proposal beyond the edge, and does it represent a breakthrough?

- NEMESIS will provide for the first time a multi-scale, quantitative, and mechanistic understanding of the effects of focal injury (stroke) on the brain. It will show that neurological deficits do not depend on local damage but distributed patterns of abnormal activity across many brain networks.
- NEMESIS will employ computational modelling in combination with animal models to test novel strategies for high precision individualized circuit stimulation to treat neurological deficits, solving the critical problem of *where* and *how* to stimulate.
- NEMESIS' circuit stimulation strategy for focal injury can be extended to other focal and possibly non-focal pathologies in which brain activity is altered (e.g., degenerative conditions, drug addiction or psychiatric disorders).
- NEMESIS will synergize neurology, neurophysiology, neurobiology, physics, computational neuroscience, and neuro-technology) to address these fundamental issues in neuroscience and health.

Why is this proposal high risk and high gain?

- The proposal is **high risk** because of the strong hypothesis that focal injuries induce structural and functional disconnection of brain networks, and that functional disconnection relates to a simple neurophysiological mechanism: sleep-like slow waves generated near the lesion that propagate and functionally impair normally connected brain regions. Moreover, we assume that this mechanism can be modulated by localized brain stimulation to restore function. Despite the high risk, our results demonstrate that focal lesions produce relatively few (low dimensional) spatial patterns of functional disconnection, therefore amenable to modelling and modulation. Moreover, our novel results since the last submission (2020) strongly strengthen the main hypothesis. However, if the central hypothesis were not correct, alternative mechanisms of disconnection can be studied and used for circuit-based stimulation.
- The proposal is **high gain** because brain disorders are extremely common, and a general strategy for circuit-based treatment has enormous potential for health and society. In Europe 164 million people, 38% of the population, suffer from some form of mental disorder, with an economic impact of 798 billion euros. Neuropsychiatric disorders are now more common than cardiovascular disorders. 25% of brain diseases are neurological with migraine, stroke, trauma, dementia, and epilepsy, the most common. All these disorders are characterized by abnormal patterns of activity in circuitries that are injured (stroke, trauma), abnormally developed (autism, epilepsy, psychiatric disorders), degenerated (Alzheimer or Parkinson's disease), or misfiring (migraine, drug addiction). Our strategy is flexible and can be adapted to other models of disconnection or circuit alteration.

Why is this proposal of high impact and synergistic?

- NEMESIS would not have been possible 5 years ago. Maurizio Corbetta, Marcello Massimini, Mavi Sanchez-Vives, and Gustavo Deco have developed in the last half decade independent bodies of work that form "**four pillars**" on which this proposal stands, and that, now and in Europe, can be synergized theoretically and experimentally to change the way we think and treat brain injuries. Furthermore, the four PIs have had previous collaborations/publications in pairs, which facilitates this synergistic project.
- NEMESIS is high impact because it connects different domains of knowledge (focal injury, sleep, brain networks) by providing a mechanistic understanding of focal lesions on whole brain activity.
- NEMESIS provides a new strategy for brain rehabilitation by harnessing different behaviourally relevant brain signals to guide circuit stimulation. This contrasts with current strategies that deal with the brain as a black box. For instance, physical therapy is applied with no regard to the underlying physiology.
- NEMESIS experiments, irrespective of the hypothesis, will create rich and deep datasets that will be used for years to come to address many other questions. For instance, our Washington University stroke database has been downloaded hundreds of times by dozens of groups for independent research.
- NEMESIS represents the synergistic endeavour of large research groups led by one neurologist (Corbetta), one physiologist (Massimini), one cellular and system neuroscientist (Sanchez-Vives), and one physicist (Deco); the combination of methods and scale of analysis: brain imaging and behaviour (Corbetta), neurophysiology (Massimini), cellular and systems neuroscience (Sanchez-Vives), and computational modelling (Deco); but, above all the integration of concepts and ideas. Overall, through the synergy of people, technology, and concepts NEMESIS provides a true "paradigm shift" for understanding and treating brain injuries.

State-of-the art: the effects of stroke on brain networks, behaviour, and recovery (1st pillar)

Since Paul Broca's first description in 1861 of speech impairment after a frontal lesion, the dominant paradigm in clinical wards and Schools of Medicine around the world has been the *neuron* doctrine. Accordingly, neurological impairment after focal injuries (e.g., stroke, trauma, epilepsy, tumours), but also degenerative (Alzheimer, frontotemporal dementia, Parkinson), or developmental (autism, schizophrenia) disorders, are thought to reflect the dysfunction of local circuitries affected by the specific pathology. For instance, in the case of stroke, motor deficits or loss of speech arise because of dysfunction of neurons specialized for motor and language functions due to loss of blood circulation with secondary tissue hypoxia. This has led to the well-established idea that lesions in different parts of the brain give rise to different syndromes; that deficit and their recovery vary across individuals, and overall that stroke is a highly heterogeneous condition. Accordingly, treatments have been developed to solve the local issue. For instance, drugs or mechanical devices are used to re-open occluded blood vessels, while exercises (rehabilitation) re-train the dysfunctional neurons. More generally, in clinical neuroscience across the board, treatments are delivered to patients while dealing with their central nervous system as a "black box". We use drugs, surgery, rehabilitation, and even non-invasive brain stimulation and measure their effect on behaviour with little or no consideration of how different treatments influence brain signals, or vice versa how brain signals influence treatment.

The **1st pillar** in NEMESIS is a new understanding of how focal lesions affect whole brain network function. It has been known for more than 100 years that focal lesions produce functional effects at a distance ("action à distance"¹ or "diaschisis"²) –the dysfunction of remote regions connected to the site of injury. With the advent of neuroimaging, diaschisis was measured as the depression of perfusion/metabolic function,³ or abnormalities of task evoked activity⁴ in apparently normal brain areas connected to the injury.⁵⁻⁷

About 10 years ago, Maurizio Corbetta and his team reported on a new pattern of diaschisis ("connectional diaschisis") consisting of the alteration of the temporal correlation of the blood oxygen level dependent (BOLD) signal measured with fMRI between brain regions, a.k.a. "functional connectivity" (FC)(Fig.2).^{8,9} This signal is a more direct marker of abnormal communication between the lesion and connected distant regions. In the last ten years, they studied prospectively and longitudinally (1-2 weeks, 3 and 12 months) hundreds of stroke patients with neurobehavioral tests and structural, diffusion, perfusion, and functional MRI. This is the largest stroke cohort in the world studied with multi-modal imaging. This body of work has established several new principles that we use in NEMESIS. **(1)** In contrast to what commonly believed, **strokes damage predominantly the subcortical nuclei and white matter, not cortex**. In fact, only 15-20% of all strokes are cortical. This means that strokes on average leave cortical regions intact but affect the

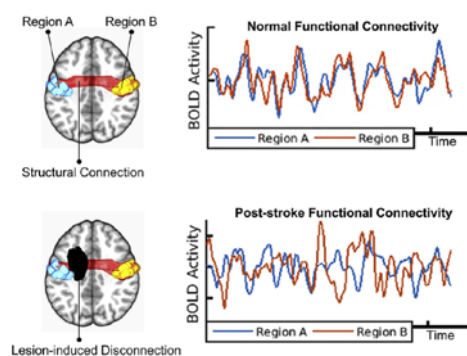


Fig.2. Connectional diaschisis. Top: normal fMRI functional connectivity (FC) between homologous regions in the two hemispheres. Bottom: stroke induces a structural disconnection with secondary abnormality of temporal correlation of fMRI signals.

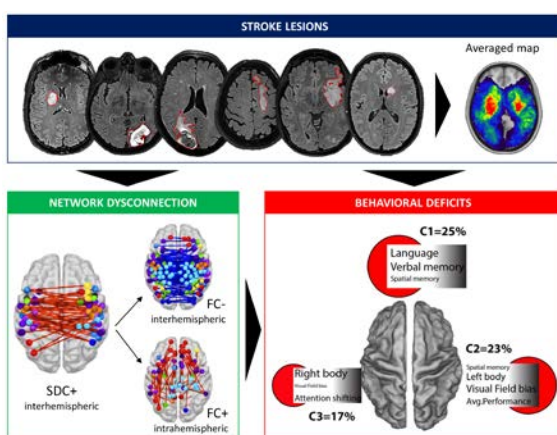


Fig.3. Top: multiple stroke lesions and average map (n=132). Bottom left: lesions cause structural (SCD) and functional disconnection (FCD): decrements (FC-) of inter-hemispheric and increments (FC+) of intra-hemispheric functional connectivity. Bottom right: lesions and SCD/FCD cause three sets of correlated deficits across domains that explain variance across subjects (e.g., C1 explains 25% variability).

communication among them through white matter pathways (Fig.2-3).

(2) At the population level, strokes do not cause many different syndromes as taught in medical schools, but **a few clusters or components of correlated deficits across domains** (e.g., language and memory, Fig.3).¹⁰ These components explain large amounts of individual variability (50-70%), persist at the chronic stage, and track recovery of function.¹¹ Thus, they represent robust phenotypes for intervention studies. **(3)** The behavioural correlation among deficits has a structure-function counterpart at the level of brain networks. Irrespective of location and volume, **strokes cause a finite number of structural (SCD) and functional connectivity (FCD) disconnection patterns that predict deficits and recovery**. Two correlated SCD-FCD patterns explain 60% of all disconnection patterns in 100 patients (Fig.3).¹²⁻¹⁴ This occurs because the same white matter tracts connect many cortical regions. Importantly, normalization of functional connectivity abnormalities at 12 months correlates with good recovery.¹⁵ Therefore, even though strokes can occur in many different locations, they surprisingly tend to cause a relatively small number of disconnection effects at the whole brain level. This is good news for circuit stimulation strategies. **NEMESIS will test the**

hypothesis that normalization of connectivity patterns through stimulation will improve deficits and promote recovery. However, this goal requires a more in-depth understanding of the physiological mechanisms of disconnection.

Sleep mechanisms and the diaschisis hypothesis (2nd pillar)

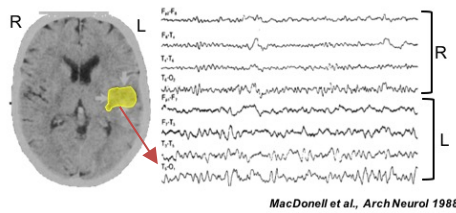


Fig.4. Slow-wave (delta) EEG activity in the left hemisphere near/at the stroke (red arrow), but also in distant regions of the L hemisphere. Some delta waves are also present in the R hemisphere.

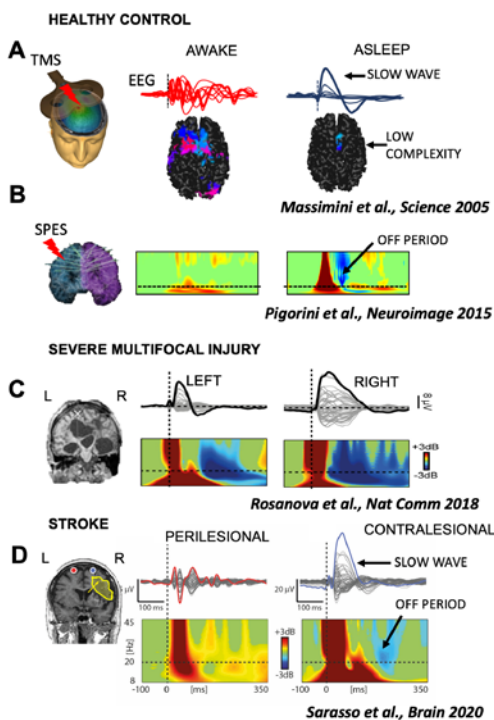


Fig.5. A. EEG responses to TMS in healthy control. Awake state: complex EEG response and network topography; Sleep: simple slow wave and low network complexity. B. Intracranial electrical stimulation (SPES) causes a complex LFP evoked response in the awake state

Old school neurologists, before the advent of CT scan, knew that focal injuries cause a slowing of the EEG signal (delta frequency 0-4 Hz) that is maximal at the location of the lesion, but that is also present in the same hemisphere, or even in the opposite hemisphere (**Fig.4**). This slowing can persist even at the chronic stage. These clinical observations were later confirmed in experimental studies in animal and human models.^{16–20} The key hypothesis of NEMESIS is that **slow-wave activity generated near the lesion spreads to directly and indirectly connected regions impairing processing. Slow wave activity intrusion through the network represents the electrophysiological correlate of diaschisis and the mechanisms responsible for functional disconnection.** This hypothesis is based on a series of observations obtained by Marcello Massimini and his team that form the 2nd pillar of NEMESIS. They developed a new approach to study changes of brain reactivity and connectivity across different brain states by employing direct cortical perturbations with TMS and EEG recordings. First, they found that in different conditions characterized by loss of consciousness, including sleep (**Fig.5A**) and anaesthesia, the **complex response to TMS normally triggered during wakefulness invariably collapses into a simple slow wave, associated with loss of network complexity.**^{21–25} Using intra-cranial electrical stimulation and recordings (**Fig.5B**), it was found that **slow waves reflect the tendency of cortical neurons to fall into a period of suppressed firing (OFF-period) after a transient increase in activity (“bi-stability”).**²⁶ Crucially, cortical bi-stable responses, by which neurons tend to “hush and forget” upon receiving an input, also occur in focal injury conditions during wakefulness, for example, brain injury patients in vegetative state²⁷(**Fig.5C**). Key for our hypothesis, in a recent study (2021) they showed sleep-like cortical responses, characterized by OFF periods and low complexity, in the perilesional cortex near a stroke (**Fig.5D**).²⁸ Furthermore, recent studies show that the intrusion

of local slow waves in a cortical region of a healthy sleep-deprived rodents²⁹ and humans³ correlate with cognitive deficits specific to the cortical site.³⁰ Overall, these findings support the idea that slow wave activity after injury may lead to loss of network complexity and loss of function. Importantly, as discussed later in pillar 3, bi-stability and slow-waves represent a basic default mode of cortical activity that can be engaged by physiological and pathological alterations of the of the inhibition/excitation balance in neural circuits,³¹ as well as by white matter lesions that disconnect cortex³². In conclusion, NEMESIS proposes that **the intrusion of slow waves, and the underlying cellular mechanisms, are a key mechanism of network disruption and behavioural impairment after focal injury.** Consequently, we posit that the **suppression of pathological slow waves in brain injured subjects should be facilitating information processing hence function.**

However, there is another side to the story. Slow waves during sleep by sending cortical circuits off-line exert a vital homeostatic regulation of synaptic excitability that promotes learning.³³ The loss of function that we experience every night during sleep is considered a price worth paying to have a more performant brain upon awakening.³⁴ Then, an intriguing possibility is that in the **acute post-stroke phase local slow waves may be useful to preserve or restore the state of the residual circuitry for subsequent recovery.** The presence of local slow waves may be still related to global dysfunction through anatomical or functional disconnection. However, the reduced metabolic cost of slow wave activity³⁵ may be helpful to reduce excitotoxicity after brain injury. Overall, **NEMESIS tests the hypothesis that sleep-like slow waves are the physiological mechanism of disconnection and examines their role in inhibiting information processing vs. promoting**

plasticity. Correction of these abnormalities, however, will require intimate knowledge of the underlying cellular events.

Cellular origin and modulation of sleep-like slow wave activity (3rd pillar)

A more thorough understanding of sleep-like slow waves requires a deeper dive into the cellular mechanisms to find out how they are generated, and how they can be manipulated either to dampen or promote their effect on network activity for the recovery of circuit function. Mavi Sanchez-Vives originally described the cellular and network underpinning of slow oscillations³⁶ and has been at the cutting edge of the field since, which is 3rd pillar of NEMESIS.

Slow oscillations emerge from the recurrent interaction between cortical neurons, making them a network phenomenon. During the active periods, or Up-states, neocortical neurons (both excitatory and inhibitory) are depolarized, receive barrages of synaptic inputs and fire action potentials. During the ensuing Down-states or OFF-periods, neurons remain hyperpolarized, and the synaptic activity is almost silent. This alternation resembles the “bi-stability” described above in human studies. Cortical recordings from different species agree on the leading role of infragranular layers—layer 5—in the initiation of Up states.^{36–38} During Up states, the activity expresses coherent oscillations at high frequencies in the beta (15–30 Hz) and gamma (30–90 Hz) range. **This phenomenon is scale-invariant such that slow oscillations display similar characteristics whether they are recorded from the whole cortex of a sleeping human or from a small piece of cortex in a plate.** This property reflects the dependency of slow oscillations on neural connectivity and the local activation and regeneration of the waves sequentially along the network are the basis of slow wave propagation.^{39–41}

Sanchez-Vives and colleagues have proposed that **fluctuations of Up- and Down-states -the origin of the slow waves-- are the “default” intrinsic pattern of cortical activity.**^{42,43} The cortex defaults to slow-wave activity due to changes in neuromodulation, as in sleep,^{44,45} due to changes in the excitatory/inhibitory balance as in deep anesthesia,⁴⁶ or because of structural lesions, as when a cortical gyrus is physically isolated by a cut of the underlying white matter⁴⁷ or in a cortical slab.⁴⁸ All these mechanisms are highly relevant for understanding how slow waves can be generated and then propagated after a stroke.

However, to gain insight on how cellular mechanisms impact network level phenomenology is necessary to develop comparable metrics across scales (micro, meso, macroscale). Since our last submission, we have worked hard to develop a common framework to describe changes in spontaneous activity and stimulus-evoked activity. We found that metrics of spontaneous activity (spectral exponent) and network complexity response to stimulation (perturbation complexity index, PCI⁴⁹) similarly capture across models (slice, in-vivo rodents, humans) the difference between slow wave and normal desynchronized activity regime (Fig.6).

NEMESIS plans a systematic exploration of the mechanistic basis of slow wave generation and propagation in lesions as well as different neuromodulation strategies including electrical (non-invasive and thalamic), pharmacology, photopharmacology or optogenetics, to induce a recovery of the complexity and processing capability of the cortical network.

However, the widespread topography of disconnection and the complex dynamics of brain response to stimulation represent real challenges

for any circuit-based treatment strategy. For instance, while transcranial magnetic stimulation (TMS) has been attempted clinically in human stroke, results have been disappointing likely because of the huge parameter space (duration, frequency, amplitude, location). Thus, leveraging the knowledge gained on slow-wave modulation in animal, we plan to use computational modelling to decide *where* and *how* to stimulate. Among different modelling strategies, our choice is to use whole-brain modelling that provides at the same time a compact description of cellular, mesoscopic, and network-like phenomena.

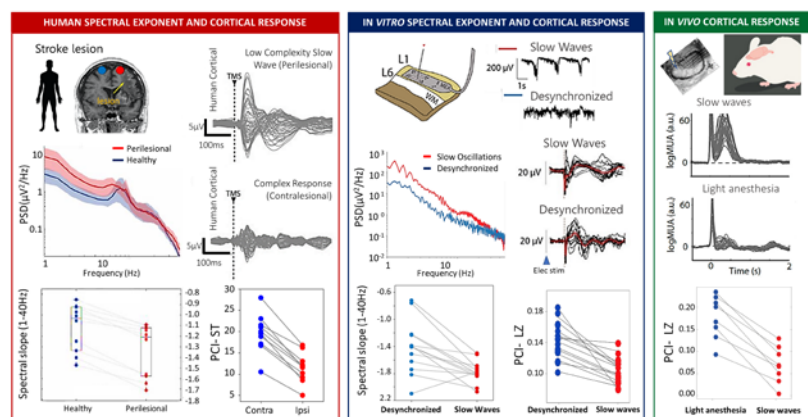


Fig. 6 Comparison metrics for in-vivo and in-vitro models. Left: Humans. A stroke causes a more negative spectral exponent (slope of the aperiodic spectrum) of EEG in the perilesional area with respect to the healthy hemisphere of stroke patients.³⁰ TMS of the perilesional area evokes a slow wave associated with a local collapse of complexity (measured by PCI), with respect to the healthy side. **Center: Cortical slice.** Cortical slices manifest spontaneously slow waves that can be desynchronized with different means. The spectral exponent quantifies the difference between slow waves and desynchronized activity. Similarly, the PCI can be used to quantify the slice network response to stimulation for slow waves vs. desynchronized activity. **Right: Rodent in-vivo model.** Increasing the level of vigilance (light anaesthesia) increases the network response complexity to stimuli (PCI) as compared to slow waves in deep anaesthesia.³⁶

Whole-brain models of healthy and lesioned brain networks (4th pillar)

Gustavo Deco is one of the pioneers of whole-brain computational modelling,^{50–52} an integrated strategy to simulate multi-scale signals and mechanisms based on a biologically realistic diagram of the anatomical connectivity coupled with models of neuronal dynamics (**Fig.7**). This is the **4th pillar of NEMESIS**.

The link between anatomical structure and functional dynamics, introduced more than a decade ago⁵³ is at the heart of whole-brain network models. The ***anatomy is represented by the structural connectivity (SC) of an individual or average brain, measured in vivo by diffusion weighted imaging (DWI) combined with probabilistic tractography.***^{54–57} The spatial resolution is in the order of 1-2 mm, but with ultra-high field MRI resolutions 0.4 mm can be reached (Padova will install a 7T MRI system by 2024). The structural connectome describes the wire-diagram of the connections between cortical and subcortical regions as seen by tractography. The ***functional global dynamics result from the mutual interactions of local node dynamics coupled through the underlying empirical anatomical SC matrix.*** Whole-brain models aim to balance between complexity and realism to describe the most important features of the brain *in vivo*. The most successful whole-brain computational models have taken their lead from statistical physics where it has been shown that macroscopic physical systems obey laws that are independent of their mesoscopic constituents. The emerging collective macroscopic behaviour of brain models has been shown to depend only weakly on individual neurons.⁵⁸

This theoretical framework has been successful in explaining the pattern of inter-regional activity correlation measured with fMRI, so called resting-state-networks^{50,51,59,60} as part of a collaborative FP7 grant

called Brain-Synch in which Gustavo Deco and Maurizio Corbetta collaborated. This platform is now accessible to a larger community through *The Virtual Brain* simulator.⁶¹ Deco and Corbetta have used ***whole-brain models to estimate the effect of stroke lesions on global network properties and have stimulated models of lesioned brains to estimate the variability of possible neural states***, or entropy, both at the level of individual regions and globally.^{62,63} Such thorough exploration of the parameter space of stimulation is not possible clinically since patients (or even rodents) cannot undergo so many experimental conditions. Notably, model parameters significantly correlated with deficits measured in the same patients suggesting that whole-brain models have the potential of estimating the clinical effect of disconnection. Recent developments important for NEMESIS are model descriptions not only of the static FC (averaged over all time points), but also of the temporal structure of activity fluctuations, the so-called functional connectivity dynamics (FCD).⁶⁴ This could be important to measure the propagation of slow waves across cortex. Also, the demonstration that recursive stimulation of whole-brain models can force transitions between sleep and wakefulness patterns of activity.⁶⁵ This is a strategy potentially applicable to our project.

NEMESIS plans to enhance whole-brain modelling in several novel ways to guide where and how to stimulate: (1) by incorporating cellular and mesoscopic signals characteristics of disconnection; (2) by validating the patterns of disconnection and slow-wave activity in individual patients, i.e., a true generative model; 3) by simulating the effect of different loci and type of stimulation to recover the best complexity and variability of neural states, as a guide toward novel treatment protocols.

Animal models of stroke

NEMESIS envisages experiments in murine models of brain lesions in parallel to those in humans. A stroke can be induced experimentally by occlusion of a blood vessel or photo-thrombosis of cortex. Studies on murine models have mainly concentrated on small cortical lesions while human strokes are larger and disconnect predominantly the white matter. Thus far two main types of structural/functional modification of the neuronal circuits have been reported. First, during an acute “critical period”, perilesional cortex undergoes processes of local synaptic sprouting with the re-configuration of local connections, and angiogenesis. This local re-organization in most cases is not capable alone, to support the recovery of the physiological function.^{66,67} Second, similarly to the finding in humans,^{9,12,68,69} a functional alteration emerges on the contralateral homologous region of the brain, likely via callosal disconnection, with a relative over-expression of glutamate

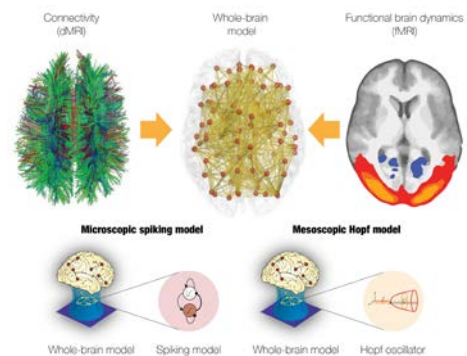


Fig.7: General Framework of Computational Modelling for Resting-state fMRI Functional Connectivity: Neuroanatomical structural connectivity (SC) data is obtained using diffusion imaging and tractography. Parcellation in N cortical areas with clear anatomical landmarks provides a matrix of strengths of inter-areal connectivity. The local neurodynamical models are coupled through the SC and used to obtain a simulated functional connectivity (FC). The optimal working point of the model is found by comparing the simulated FC with the empirical FC as a global coupling parameter is varied. Examples of two different type of local dynamics are shown (bottom): left) microscopic biophysically realistic networks of excitatory and inhibitory neuronal populations; right) mesoscopic Hopf bifurcation model for describing oscillatory behaviour and entrainment.

receptors and excessive response to stimuli and tasks. The notion of over-activity in the contralateral hemisphere, with consequent excessive interhemispheric inhibition on perilesional cortex (push-pull), has guided most efforts, but as earlier reviewed human strokes cause widespread patterns of disconnection. Currently network-wide abnormalities in murine subcortical strokes are largely unknown.^{70,71}

A stroke model in living rodents represents the only way to link the cellular mechanisms underlying the intrusion of sleep-like slow-waves activity to structural-functional disconnection patterns of humans. Functional imaging at the population and brain-wide scale in combination with electrophysiology and optogenetics can address these open questions. Moreover, rodent models will be fundamental for a quantitative evaluation of the rehabilitative effects of interfering with slow wave dynamics. Rehabilitation re-organizes horizontal and thalamo-cortical connections with a functional remapping in neighbouring cortex, re-tuning of functional properties of receptive fields and re-focusing of cortical patterns toward a normal condition.⁷² Researchers at UNIPD have successfully modulated recovery of motor function^{72–74} with a series of manipulation including robotic rehabilitation, optogenetic stimulation, and other neuro-modulation strategies.

Rehabilitation and neurostimulation approaches for recovery of function

Traditionally rehabilitation has employed task-driven exercises, e.g., repetition of skilled motor tasks, to improve recovery function after injury with weak-moderate efficacy.⁷⁵ Rehabilitation is applied with no regard to the underlying lesion characteristics and even less brain signals. More recently, there has been great interest in non-invasive brain stimulation (magnetic or electrical). However, a recent Cochrane analysis showed that **repetitive TMS (rTMS) was not associated with a significant increase of activities of daily living or motor function**.⁷⁶ Similarly, another Cochrane review on electrical stimulation (tDCS) did not find any effect on life skills, and a modest effect on upper limb function at the end of the intervention, but not 3 months later.⁷⁷ This dearth of effects reflects lack of understanding of the underlying mechanisms. There have been some promising results with closed-loop stimulation,^{78,79} both invasively,^{80,81} and non-invasively.^{82,83} There have been also recent technological developments in multi-site TMS^{84–86} that could be very helpful in the context of widespread disconnection. Connect-2-Brain, an ERC Synergy award, which Massimini and Corbetta advise, proposes the development of a multi-site TMS system coupled with high-density EEG for closed loop control.

Beyond the state of the art

NEMESIS will provide the first complete characterization of the effects of focal lesions, both at the level of local circuits and brain networks, as measured with multi-modal imaging and electrophysiology (fMRI, DTI, EEG, multiphoton and mesoscale calcium imaging, and high-density probes) in multiple systems (human patients, rodent lesions and stroke model, and local circuits in slice). NEMESIS will clarify the cellular and circuit mechanisms of network disconnection after focal injury. It will also characterize the effects of disconnection on the responsiveness of brain networks. It will define the conditions and the stimulation paradigms through which it is possible to rescue abnormal dynamics in affected circuits and restore the normal complexity of neuronal responses and network states. NEMESIS will develop generative whole-brain computational models by incorporating cellular and mesoscopic signals, in addition to classic structural macroscopic signals, and guided by animal models, simulate the effect of different treatment protocols in terms of location and type of stimulation. Such models will guide proof-of-concept interventions that combine circuit stimulation and behavioural training.

Objectives

NEMESIS has been organized in a set of high level inter-disciplinary objectives, each including multi-scale analysis that require the synergy of the four investigators. Each objective will be reached through a series of tasks that will generate the necessary information to move the project forward step-by-step toward the goals set out earlier.

- **Objective 1 (O1):** to link post-stroke functional-structural disconnection with local and global sleep-like brain dynamics in different lesion models (stroke, thermocoagulation) and systems (humans, rodents, slice).
- **Objective 2 (O2):** to investigate and correct the mechanisms through which sleep-like slow-wave activity biases network activity at the whole brain level (human, rodents) and in slices.
- **Objective 3 (O3):** to develop predictive whole-brain computer models of individual patients or rodents with stroke; to simulate different stimulation protocols with the goal of normalizing abnormal local and global dynamics; and, to use this information for optimizing treatment protocols that combine stimulation with behavioural training.

Section b. Methodology

Objective 1: to link post-stroke disconnection with local and global sleep-like brain dynamics

Task 1. Disconnection, slow waves, and behaviour in humans (Corbetta, Massimini, Deco)

Aim: To relate the patterns of lesion-induced disconnection measured with MRI with lesion-induced slow waves measured with EEG, we will measure simultaneously MRI and EEG signals in first time stroke patients sub-acutely (1-2 weeks) and chronically at 3 months, the period corresponding to the maximum behavioural recovery.¹¹ This study will identify whether the topography of structural (SCD) and functional (FCD) disconnection matches the topography of slow wave activity, or whether alternative signals explain connectional diaschisis. Our strategy for circuit stimulation does not strictly depend on the slow wave hypothesis. This study will also establish the most sensitive signals (SCD, FCD, EEG), or combination thereof, to track behavioural deficits and recovery. A behaviourally relevant biomarker is essential for circuit-based stimulation, and EEG signals would be much easier to obtain than MRI based signals. A scenario characterized by slow wave activity persistence at the chronic stage in patients who do not recover is consistent with a detrimental role of slow waves in information processing. In contrast, a positive relationship between acute slow-wave activity and recovery would be consistent with a role in promoting plasticity. The two above scenarios are not mutually exclusive as it is conceivable that post-injury slow waves may be initially protective (in the acute phase) but then detrimental if they persist indefinitely (throughout the chronic phase).

Methods: N=60 first time stroke patients (age 18-90) with acute neurological impairment and any lesion location, will be studied at 1-2 weeks and 3 months with: Neuropsychology: Oxford Cognitive Screen, NIH stroke scale, Stroke Impact Scale, and NIH toolbox. We will extract the component deficits identified in previous studies to track deficits, recovery, and outcome. Structural MRI: lesions will be segmented on T1/FLAIR sequences. Diffusion MRI: multi-directional and multi-weighted diffusion weighted images. Probabilistic tractography will be run with spherical deconvolution to connect regions of interest based on R-fMRI. Tract-specific measurements: volume, fractional anisotropy (FA) (9, 56), and Hindrance Modulated Orientational Anisotropy (57). Resting state fMRI: TE = 27 ms, flip angle = 90°, 3x3x3 mm isotropic voxels, volume TR = 2.00; pairwise normalized Pearson values from the Schaefer-Yeo atlas parcellation and subcortical nuclei after pre-processing including motion scrubbing. High-density EEG (hd-EEG)_{nsi}: 128-channels, frequency bands: 0.01-100 Hz. In a subset of subjects, hd-EEG alone will be also acquired during sleep to assess changes in the overnight temporal dynamics of slow wave activity. As control group we will study N=30 age-education matched controls, typically family members of our patients.

Analysis: Structural-functional map of (altered) connectivity will be generated in each subject^{12,87}. Hd-EEG data will be reconstructed in source space using cortical parcellation and thickness to constrain solutions.⁸⁸ The presence of slow waves will be quantified at the source level in the time domain (period-amplitude analysis) and frequency domain (spectral analysis and slope of the aperiodic components of the spectrum, i.e., the spectral exponent). Based on the hypothesis that slow waves increase near the stroke and in connected areas, we will generate independent maps of SCD, FCD, EEG slow waves and compare them. We can also look for the EEG patterns that best explain structure-function disconnection using parcel-wise regression analysis. These signals (SCD, FCD, EEG) can be input to machine learning models to predict behaviour and recovery. Feasibility is established by new (October 2021) pilot studies (**Fig.8**).

Extension 7T MRI: The University of Padova will acquire in 2024 a new generation 7T MRI scanner, the first in Southern Europe. 7T enhances: 1) the visualization of anatomical structures and connections (resolution of 0.2-0.4 mm comparable to ex-vivo resolution); 2) functional mapping of cortical layers; 3) temporal resolution of fMRI signals (600-800 ms). We plan to scan a few selected cases of ‘canonical’ strokes, e.g., cortical vs. white matter vs. cerebellum, to generate state-of-the-art maps of disconnection and whole brain models. By including separate parameters for laminar function, high resolution white matter diffusion mapping, cortical variability of cellular content for each region, and slow waves, we expect to improve the accuracy of whole brain model prediction (Task 6). While 7T technology enhances novelty, the project does not depend on it.

Outcomes: (1) Single subject maps of SCD/FCD/EEG disconnection; (2) Statistical models relating behavioural deficits to SCD/FCD/slow wave topography; (3) SCD/FCD/EEG as inputs to brain models

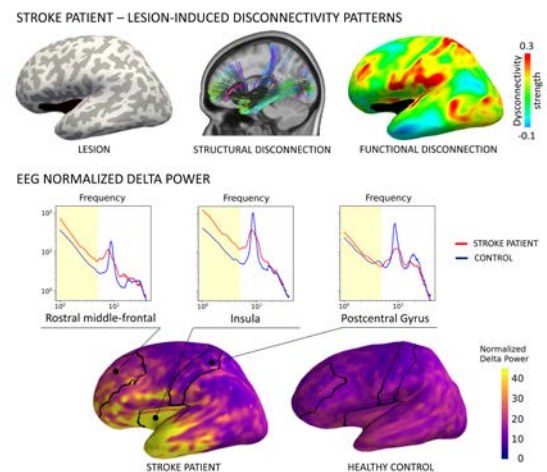


Fig.8: Left inferior frontal stroke (a) with related pattern of diffusion SCD (b), fMRI FCD (c), and EEG delta activity (e) (d). Hd-EEG signals were analysed in source space independently from fMRI to generate maps of ‘delta activity power’. The small insets (d) show the power spectra from several regions. Note increased delta power near/at lesion and distant sites, some overlapping with fMRI disconnection.

Task 2. Thermo-coagulation, connectivity, and slow-wave activity in humans (Massimini, Corbetta, Deco)

Aim: While the previous task focuses on whole brain (macroscale) non-invasive signals (fMRI, EEG) it is essential to understand physiological mechanisms at the mesoscale. In very recently work Massimini has shown that Radiofrequency-Thermocoagulation (RFTC) of a cortical site in epileptic patients causes the onset of slow waves measured as local field potential intracranially with stereo-EEG (sEEG). Slow waves were found near the lesion but were also detected in remote cortical sites.⁸⁹ It is unknown whether the topography of slow waves matches that of disconnection. Also, whether remote slow waves reflect direct anatomical disconnection or a physiological “spread” of this pattern through normal connections. We plan to address this question by combining intracranial sEEG, RFTC,⁹⁰ and advanced neuroimaging. This analysis is much more spatially resolved than the one possible with fMRI/EEG (Task 1). The use of local field potentials also affords more accurate analysis of the spread and directionality between cortical sites.

Methods/Analysis: sEEG spontaneous activity will be recorded both before and after the induction of RFTC lesions in 30 pharmaco-resistant epilepsy. LFP activity will be analysed from up to 150 contacts located both near and far from the lesion site. Recording sites that are functionally connected to the site of lesion will be identified by cortico-cortical evoked potentials elicited by single-pulse electrical stimulation performed before lesioning, as well as by DTI tractography and fMRI connectivity. The network effects will be also assessed by simultaneous hd-EEG scalp recordings to link meso to macroscale. The presence of post-lesion slow waves and OFF-periods will be assessed using time- and frequency-domain methods both during wakefulness and sleep. The topography of slow waves, their phase-relationships and directionality will be compared with maps of disconnection based on DTI and fMRI. **Fig.9** shows recent preliminary results (October 2021) supporting the main hypothesis that disconnected regions far from the lesion show slow waves.

Outcomes: (1) Directional maps of slow waves (LFP) and functional/structural disconnection; (2) SCD/FCD/LFPs as input to whole brain models. We will distinguish between direct anatomical from indirect functional disconnection. The mesoscale description is important for linking non-invasive human measurements (Task 1) with microscale rodent models (Task 3). It is also important to integrate slow waves in brain models. Alternative signals can be incorporated if slow waves and disconnection do not match.

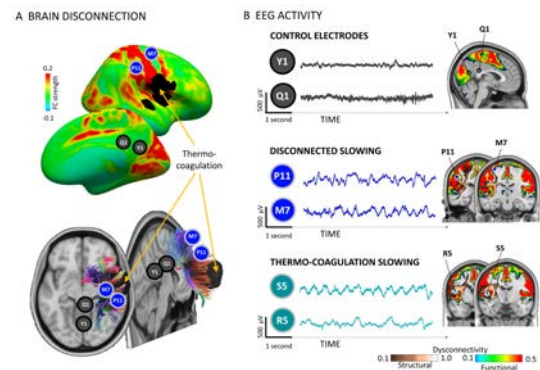


Fig.9. Single subject RFTC experiments combining DTI, fMRI, and sEEG. The lesion (black) causes a pattern of SCD and FCD. sEEG electrodes corresponding to the region of disconnection in precentral (M7) and post-central gyrus (P11) manifest slow waves, whereas electrodes in medial parietal cortex (Y1, Q1) not showing SCD/FCD show normal activity. Slow waves are also present at the site of RFTC (S5, R5).

Task 3. Stroke rodent model: disconnection and slow-wave activity (Corbetta, Sanchez-Vives, Deco)

Aim: The main goal is to develop a validated murine model of human stroke. First, we plan to develop and contrast cortical vs. subcortical models of stroke in terms of behaviour. Next, longitudinal electrophysiological and optical recordings from the perilesional and contralateral sides will allow to describe local activity and inter-hemispheric coupling during the entire post-stroke recovery phase. Third, we will perform whole brain optical imaging to map the spatial features of the slow wave topography. Fourth, these correlation approaches will be integrated with targeted optogenetic stimulation to compare maps of intra- and inter-hemispheric activity and connectivity in resting and stimulated conditions.

Methods/Analysis: Stroke in adult mice will be induced either via middle cerebral artery occlusion (MCAO)(cortico-subcortical model) or via injection of the powerful vasoconstrictor, N5-(1-Iminoethyl)-l-ornithine (L-NIO) via a micropipette into the white matter, at three locations in rostral areas (pure subcortical model).⁹¹ A battery of behavioural tests (e.g., grid walk test, Schallert cylinder) will be carried out before (baseline) and at different times after the stroke to comprehensively evaluate motor function. We will build a multiparametric motor score describing behavioural performance at different times after the lesion, to stratify ischemic mice based on “good” and “poor” outcome. Optical recordings will be carried out using mice expressing a fluorescent Ca^{2+} -based reporter of neuronal activity from the family of GCaMP6/7 molecules (GP5.5 Dkim/J. mice or via infection with AAVs). Based on this, brain wide activity over a field of view of $20 \times 20 \text{ mm}^2$ will be recorded at high frame rate with a CMOS camera, to reconstruct the patterns of neuronal activity in the proximity of the lesion, in perilesional cortex, and in the rest of the cortical surface. A seed-based analysis of the correlation patterns will provide functional connectivity maps. The whole brain optical imaging studies will confirm the widespread pattern of disconnection and slow waves found in human subjects, specifically the low dimensionality of disconnection patterns and their behavioural relevance (**Fig.3**). A static resting state analysis will be complemented with dynamic and directional analysis of effective

connectivity. For this, we will integrate the mesoscale imaging system with optogenetic stimulation relying on light sensitive modulators of neuronal activity in a combination with targeted illumination techniques.^{92–94} The directionality of effects is very important. Do inter-hemispheric alterations of connectivity primary or secondary to intra-hemispheric alterations (Fig.3)? This analysis will be complemented with longitudinal population activity recording by means of multiphoton imaging to map at high resolution in three dimensions the neuronal dynamics associated with slow wave activity following the induction of the stroke.^{95,96} Neurophysiological recordings in the deeper structures will be conducted by implanting chronic electrodes in the two hemispheres, and recording LFPs longitudinally, before and at different times after stroke. Our preliminary experiments confirm a loss of inter-hemispheric activity correlation similarly to what seen in humans (Fig.10), and an abnormal stimulus-evoked response from the lesion vs. normal hemisphere consistent with reduced function in the perilesional cortex (Fig.10). Interestingly, the loss of correlation occurs at higher frequencies (gamma) that could also be target of intervention.

Outcomes: (1) whole brain maps of calcium imaging activity in a clinically relevant murine stroke model; (2) associated electrophysiological multi-site bilateral recordings; (3) directional maps of interactions and slow wave propagation; (4) maps of local excitability and network response based on stimulation; (5) LFP spontaneous and evoked activity as input to whole brain models.

Objective 2 (O2): to investigate the mechanisms through which local injury biases network activity at the whole brain level (human, rodents) and in cortical slices.

Task 4. Human cortical stimulation of stroke networks (Corbetta, Massimini).

Aim: While task 3 develops a mouse model of human stroke, and employs cortical stimulation to assess network alterations from a causal perspective, task 4 brings this perturbational approach to human patients. We will test the idea that that the generation and propagation of slow waves in perilesional and remote portion of the cortex after focal injury are associated with an impairment of cortico-cortical information transmission and complexity, as measured by the dynamics of the EEG response to a direct perturbation with TMS. We plan to systematically stimulate with single pulse TMS perilesional cortex, contralateral homologous cortex, and other cortical regions that are directly or indirectly disconnected from the lesion, or that show or not slow wave activity (Fig.8). Hd-EEG signals are recorded from the whole brain, and we plan to use metrics of spontaneous activity and stimulus-evoked activity recently validated across scales (spectral exponent and PCI, Fig.6). TMS will reveal local bi-stable responses and global reduced network response complexity not only near the lesion, but also in remote regions directly or indirectly connected to the lesion (preliminary results Fig.9). Further, we will test the causal link with behavioural impairment by performing longitudinal measurements in sub-acute and chronic patients. In patients in which abnormal stimulus-evoked responses persist chronically recovery will be less complete. The alternative hypothesis is that the presence of slow waves and abnormal stimulus-evoked responses, acutely, represent a mechanism of homeostatic regulation of synaptic excitability.³³ Acute phase abnormalities may then correlate with chronic functional recovery.

Methods/Analysis: A new group of 60 stroke participants will undergo longitudinal assessments (1-2 weeks and 3 months) by means of combined TMS and hd-EEG recordings. We will obtain structural, diffusion, functional MRI to map the lesion and related structural/functional disconnection at the individual subject level. The relative maps of disconnection will be uploaded on the neuro-navigation system from which 5 sets of coordinates will be computed: perilesional, contralateral homologous, directly disconnected (n=1 connection from the lesion), indirectly disconnected (n>2 connections away from the lesion). TMS intensity will be based on the estimated-induced electric field on the targeted cortical surface. We will use TMS intensity of approximately 110-120 V/m, which in previous TMS-EEG studies has been shown to be effective in eliciting EEG response in adults,²⁷ as well as sham stimulation to control for unwanted sensory activations.⁹⁷ Each participant will undergo 200 TMS trials with inter-stimulus intervals randomized between 1.5–1.7 s. The whole session will last 1.5 hours, of which 60 min set up and 30 min stimulation. EEG will be recorded with a 124-channel TMS-compatible amplifier (Bittium NeurOne, Kuopio, Finland). We will compute time-frequency plots, connectivity metrics based on source analysis and local and global complexity measures from the EEG

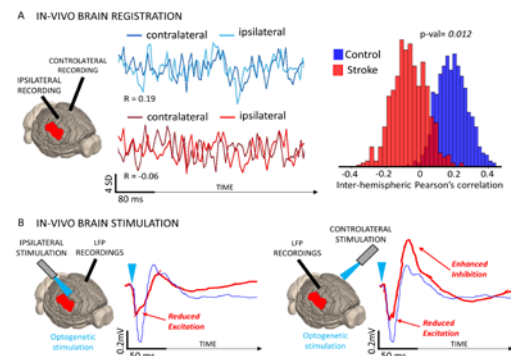


Fig.10. A) Broadband activity time course recorded from two homologous sites in healthy vs. cortical stroke mouse. Note loss of interhemispheric correlation in stroke. B) Optogenetic stimulation from either stroke or normal hemisphere, and contralateral recordings. Left: Reduced excitation in the normal hemisphere after stimulation of stroke hemisphere. Right: In contrast, stimulation of the normal hemisphere produces enhanced inhibition and reduced excitation in the stroke hemisphere. The blue trace is a healthy control.

responses elicited from each TMS targeting site.^{24,27,49} These perturbation-based electrophysiological indices of network interactions will be computed and compared in disconnected vs. non-disconnected regions of the brain, and compared to those obtained in 30 healthy controls stimulated in the same areas. The longitudinal assessment of perturbational indices of bi-stability, effective connectivity and complexity will be then correlated to behavioural recovery.

Outcomes: This study will provide (1) a map of the effect of stimulation from sites that show direct vs. indirect disconnection on local/global network responses; (2) a statistical model relating local/global network responses to stimulation and acute behavioural impairment/recovery. (3) Spontaneous and stimulus-evoked parameters (e.g., PCI) as input to whole brain models.

Task 5. Slow waves generation, propagation, and neuromodulation in vivo and in vitro animal models (Sanchez-Vives, Corbetta, Massimini and Deco)

Aim: This task will study the cellular mechanisms of lesion-induced slow waves, their propagation, and resulting dysfunction on network responses. Moreover, our *in-vivo* and *in-vitro* models will allow us to explore using the same analytic framework (**Fig.6**) how neuromodulation of local patterns of activity (including activation, silencing or frequency entraining), affects the functional global network properties, including short and long range ipsilateral and contralateral cortex.⁹⁸ This information will be also valuable for model integration and for the design of strategies to revert functionality to the network.

Methods/Analysis: Studies will use a cortical slice preparation from either animal or human tissue and can include interhemispheric connections,⁹⁹ or bilateral recordings *in vivo* to consider callosal disconnection. The first set of experiments will manipulate activity without inducing lesions to explore local vs. global impact. To locally silence activity, we can use different strategies. For instance, preliminary findings indicate that negative DC fields are effective (**Fig. 11**). We will design specific probes of electrodes or graphene microtransistors in collaboration with the CNM in Barcelona (we have 2 previous patent applications in collaborative developments) for high resolution spatiotemporal control of activity. We can also apply locally agents like muscimol, TTX, botulinum toxin. Further, we will also use a new photo-pharmacological drug, e.g., Fulgazepam a GABA_A agonist¹⁰⁰ that can be controlled by light with high spatiotemporal resolution and have some advantages over optogenetics (no genetic manipulation needed). For local activation, we will use positive electric fields (**Fig.11**), activating drugs (glutamate, ACh, norepinephrine¹⁰¹ -agents that induce specific high frequency patterns), and photo-drugs such as PAI, (muscarinic agonist, not yet commercial that we have recently characterized¹⁰²), or GABA_A antagonist Azo-Nz1⁹¹. The long-range impact of local enhanced synchrony and entrainment, as well as desynchronization can be investigated not only chemically and modulating the excitation/inhibition balance, but also by electric fields and by thalamic stimulation¹⁰³ (**Fig.11**).

For the use of novel photo-drugs we have an ongoing collaboration with Dr. Gorostiza at IBEC in Barcelona. Then, we will move on towards the local changes in activity induced by lesion-induced thermocoagulation, comparably to Task 2 in humans, but now under experimental control of both location and intensity. The focus of our analysis will be specifically on the effect of the lesion-induced slow waves and how those can be modulated: enhanced, frequency-modulated or abolished (**Fig.11**) to provide mechanism-based strategies for interventions in humans. Our ipsilateral and contralateral recordings will capture the multiscale dynamics of activity by combining measures comparable to those recorded in humans (e.g., LFPs with arrays of electrodes) with those non-attainable in humans, such as network calcium imaging, recordings from smaller circuits, single units, or even intracellular recordings. Human cortical slices will facilitate translation of results. *In vivo*, and to bridge with fMRI/EEG/TMS recordings in humans, we will use a non-contact, wide-field technique, LSCI,¹⁰⁴ that enables imaging of variations of cerebral blood flow and oxygenation of haemoglobin. Importantly, the most effective strategy identified *in vitro* for interfering with lesion-induced slow waves will be brought back to the *in-vivo* MCAO stroke model. We will employ LFPs and mesoscale calcium imaging (as in Task 3) to test for suppression of sleep-like slow waves and re-establishment of functional connectivity, respectively, following appropriate stimulation. We will not necessarily use all the techniques simultaneously, but we have experience with all of them, as well as with multilaminar in depth arrays, which we can use in

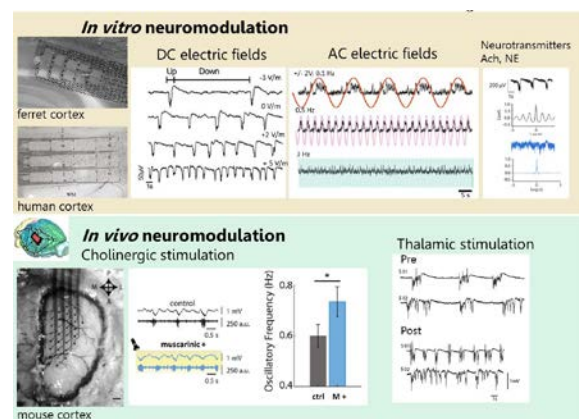


Fig.11. Top: Cortical slices express slow-wave activity that can be desynchronized with DC, AC electric fields or neurotransmitters. Bottom: *in-vivo* high-density electrophysiology. Slow-wave activity can be desynchronized through cholinergic stimulation or thalamic stimulation. All results from Sanchez-Vives lab.

combination. In the last years we have developed novel analytical network measures including spectral properties, wave propagation, functional complexity, perturbational complexity index in vivo/in vitro, singular value decomposition, and others to quantify cortical network dynamics. Some measures, we have recently shown (**Fig.6**) cover the span of spatial scales. We will thus obtain data comparable to that from humans useful for integration and validation of computational models.

Outcomes: These experiments will provide a mechanistic insight into the impact of local activation and inhibition near the site of stimulation (perilesional) as well as in remote (also contralateral) areas. The lesion experiments will evaluate how best to modulate lesion-induced slow waves and their impact on network dynamics.

Objective 3 (O3): to develop predictive whole-brain computer models of individual patients or rodents with stroke; simulate different stimulation protocols with the goal of normalizing abnormal local and global dynamics; and use this information for optimizing protocols that combine stimulation with rehabilitation.

Task 6. Whole-brain models of disconnection and slow-waves (Deco, Corbetta, Massimini, Sanchez-Vivez).

Aim: The goal is to use whole-brain models to predict the effect of lesions on brain networks in single individuals (human, rodent), and in different experimental paradigms (stroke, thermocoagulation). In addition, by introducing data from EEG or LFPs on slow waves, we will characterize the dynamics of slow waves, and how they propagate from the perilesional region to the rest of the network.

Methods/Analysis: The whole-brain human model combines structural neuroanatomical data (diffusion tensor imaging) and brain activity data (fMRI, hd-EEG, LFP). The anatomical connectivity is given in the form of density of axonal fibers according to a certain parcellation scheme. The global dynamics results from the mutual interactions of local node dynamics coupled through the underlying structural connectivity matrix. We

plan to use two different approximations of the local dynamics, namely, a biophysical realistic mean-field reduction of conductance model for spiking neurons,^{52,105} and a mesoscopic Hopf model, useful reduction for emulating oscillatory local dynamics (**Fig.7**).^{64,106} The mathematical formulation of the biophysical realistic dynamical mean field model and Hopf model are in.^{64,107} This approach yields a series of static and dynamic parameters, e.g., *metastability*, *average*, and *dynamic functional connectivity*⁶⁴ that can be used to characterize the dynamic state of a normal or lesioned brain. Testing both models will allow us to verify which amount of biological detail is necessary to embed slow-wave dynamics, and their manipulation, within whole-brain models. In addition, we will estimate “brain states” with a Probabilistic Metastable Sub-states (PMS) framework as explained mathematically in Deco et al.⁶⁵ This technique captures in a succinct manner the most common dynamic spatial network configurations (dynamic FC, dFC) and their temporal structure (frequency, duration, switching). These different static and dynamic parameters/states can be used to describe differences between normal and lesioned brain.^{62,63} Preliminary results show moderate correlation between fMRI-FC estimated by the model and empirically measured fMRI-FC using the static model in the same patient as Fig.8 (**Fig.12**).

However, here, we wish to implement a true generative predictive model of the effects of lesions on connectivity. (1) For the *predictive model*, the whole-brain model will be fitted to the FC matrices of individual healthy control participants and to the group-averaged FC matrix (separately, we can test Functional Connectivity Dynamics (FCD) and Probability Metastable States (PMS) fitting). Model fitting will be accomplished by varying both the global coupling parameter and effective SC parameters. The predicted FC matrix for each patient will then be generated by lesioning the fixed-parameter healthy control model via the individualized disconnection structure of each patient. We will induce sleep-like slow-waves in the damaged local regions by modifying the underlying conductivities (AMPA, NMDA and GABA) and rate adaptation level.¹⁰⁸ This model will have several improvements over previous models available. First, we will estimate the stroke damage based on structural connectivity disconnection, not only cortical damage as in previous studies.^{109,110} Corbetta and his team have recently shown that SC disconnection is a much better prediction of FC alterations than lesion damage¹³: directly embedding SC disconnection in the models should greatly

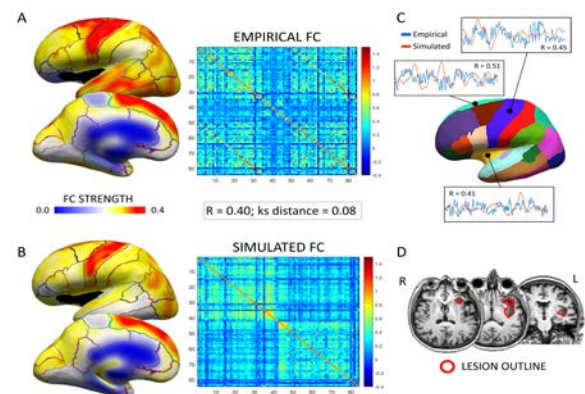


Fig.12. Personalized whole brain model of the same stroke patient in Fig.8 using the patient's own diffusion structural connectivity and Hopf oscillators. The functional connectivity map of the model (FC sim) significantly resembles the empirically measured FC (FC emp). FC strength is the mean FC between that parcel and the rest of the brain. Model timeseries show a moderate correlation (0.4-0.5) with empirical fMRI timeseries.

increase their ability to estimate FC abnormalities. As in Griffis et al.¹³, we will embed the patient's segmented lesion into a healthy control structural connectome and determine the proportion of streamlines between each pair of regions that are interrupted by the lesion. This atlas-based procedure does not require patient-specific tracking of structural connections, but only the segmentation of the lesion that can be obtained through clinical scans. It can be validated with direct measures of SC disconnection based on diffusion imaging. Second, we will investigate the effect of slow-wave generation in the lesioned region and their propagation across the pathways of disconnection, to establish a mechanistic relationship between lesions, disconnection, and slow-wave dynamics. (2) For *modelling the effect of thermocoagulation*, we will consider patient-specific whole-brain models built from his/her own structural connectivity, lesion them at the location of the thermocoagulation, and induce sleep-like slow-waves. The model results will be directly compared with SEEG data empirically measured before/after coagulation. (3) The *rodent whole-brain model* will be constructed in similar ways but using detailed anatomy tracing studies (Allen Structural Connectome of mice) and fitting the generative parameters of the model (dynamical parameters and underlying effective connectivity) with the empirical calcium-based activity data and multisite neurophysiological recordings (spike activity and LFP). This animal model bridges non-invasive (fMRI, EEG) and invasive human data (SEEG) with neuronal activity and circuit level models.

Outcomes: A generative whole-brain model framework will offer a detailed causal mechanistic explanation of the relation between stroke damage and its propagation. A full characterization of the patterns of functional disconnection and sleep-like slow-wave dynamics is fundamental for understanding strategies for diagnosis, prediction, and most importantly stimulation-based recovery. These benefits will be further enhanced by high spatial (0.2-0.4 mm) and temporal (600-800 ms) resolution 7T MRI studies in Padova.

Task 7. Whole-brain models on the effect of external stimulation (Deco, Corbetta, Massimini, Sanchez-Vives)

Aim: Here we plan to use whole-brain models to simulate the effects of stimulation in lesioned systems. We will systematically vary the location and number of sites of stimulation, as well as the type of stimulation (excitatory, inhibitory, noise). This modelling is necessary for solving the problem of *where* and *how* to stimulate to facilitate the normalization of patterns of altered connectivity and dynamics.

Methods/Analysis: Individual whole-brain models will be built for each patient, as described above. Static FC, dynamic FC, and PMS will be measured, as well as their distance from a distribution of healthy parameters. In addition, we will compute parameters derived from stimulation on the complexity of the evoked response, i.e., PCI. These parameters will be estimated from models trained on different signals: fMRI, hd-EEG, SEEG. In rodents, we will estimate calcium imaging and LFPs. We plan to use two 'spatial' (*where*), and two 'temporal' strategies (*how*). The first spatial criterion builds on our recently proposed scheme to facilitate transitions from sleep to awake state⁶⁵ (**Fig.13**). Having reduced dynamic anomalies to a few parameters, such as PMS frequencies, one can recursively find the optimal stimulation loci (individual nodes or a combination of nodes) that, when stimulated with external currents, allow reverting the anomalies. The second spatial criterion will take advantage of the low dimensionality of the disconnection patterns as shown in humans. Computing a spatial PCA on the regional disconnection, we will select for each individual subject brain the most representative sites and verify whether stimulation of these sites allows to manipulate the PMS. The PCA can be computed for example on the fMRI data, and the PMS can be read out on the EEG data, which are easier to monitor dynamically during rehabilitation. In terms of the temporal strategy, we will first try to directly manipulate local slow-wave activity, increasing or dampening it, and verify, accordingly, the normalization or worsening of connectivity or states. An alternative approach will act on slow-wave activity indirectly: we will inject peri-lesionally, network-wide through a spatial PCA, or more globally, e.g., thalamo-cortical connections, either noise or high frequency signals (gamma activity). The rationale here is that sleep-like slow waves represent a powerful attractor state that needs to be overcome. Increasing signals in pathways that are necessary for a particular task, or increasing noise diffusely are two possible strategies for moving the attractor landscape. We see the modelling and animal work on stimulation informing each other in the development of an optimal stimulation + rehabilitation intervention to treat specific deficits. In the human case, stimulation in-silico will consider the distribution of electrical and magnetic fields through the skull onto the brain by explicit models solving the Maxwell equations in individuals. Finally, the

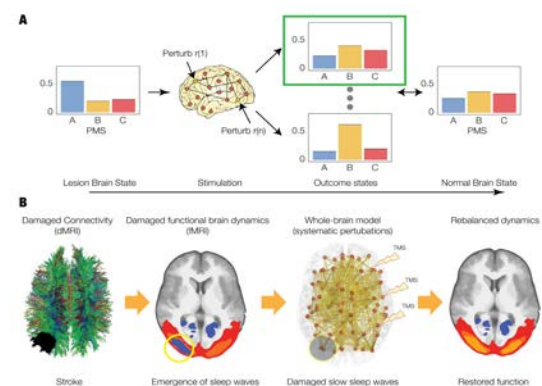


Fig.13. Whole-brain models for stimulation. Recursive stimulation of single nodes or combinations a set of features (states) is obtained on each iteration. The goal is to find patterns that decrease the distance with a normal state.

normalization induced by stimulation will be transient. A possible development would be to implement in the model a plasticity mechanism to study the effect of repeated sessions of stimulation. Another development would be to use task data to approximate a desired task state, which could be then reached through repeated stimulation and a closed-loop system. Some of these possibilities will be explored during the grant.

Outcomes: The use of lesioned human, rodent and slice systems will permit the exhaustive investigation of the parameter space of stimulation that can be used for forcing a transition to a different brain state.

Task 8. Model-based rodent brain stimulation during rehabilitation (Corbetta, Deco, Sanchez-Vives)

Aim: Methods of stimulation studied in task 5, and strategies of stimulation developed through modelling in tasks 6-7 will be used to develop a protocol for treatment that integrates brain stimulation with rehabilitation. The goal will be to modulate slow waves (or alternative physiological mechanisms related to disconnection) either directly or indirectly through noise or high frequency stimulation, and normalize network complexity (PMS, PCI). We will focus on the physiological mechanism related to functional/structural disconnection since in human studies disconnection strongly relates to the degree of behavioural impairment and recovery. First, we will test the effect of stimulation on-line in a motor task. Then, the treatment will be applied chronically over several sessions paired with motor rehabilitation delivered by a robotic device with which UNIPD researchers have extensive experience.⁷²⁻⁷⁴

Methods: Infarcts will be caused via MCAO in adult mice separating cortical from subcortical. Stimulation will be carried out via either optogenetic approaches or electrical micro-stimulation before or in phase with movement execution. Optogenetic neuromodulation will drive identified nodes within the whole brain network with activity oscillations at specific spectral bands. We will express the light sensitive depolarizing channel ChR2 in pyramidal neurons or fast spiking interneurons (Parvalbumin-positive) for modulating slow and fast rhythms respectively. This high-resolution neuromodulation strategy, in combination with brain wide optical recordings represent the closest approximation to the stimulation scenario evaluated with the model. A second approach is frequency-locked electrical micro-stimulation strategy that has given encouraging results in preliminary unpublished work. Mice display a dramatic reduction in skilled reaching abilities post-stroke.^{74,111} We will test the hypothesis that appropriate stimulation modulating slow waves (before or during movement execution) enhances on-line grasping precision. The same approach will be applied chronically, in daily rehabilitation sessions post-stroke, to determine whether it leads to functional gains that persist beyond the treatment period and generalize to non-trained sensory-motor tests.

Outcomes: This task will test the effectiveness of novel, model-based protocols of modulation and rehabilitation aiming at restoring connectivity and network complexity during motor training in rodents.

Task 9. Model-based human brain stimulation during rehabilitation (Corbetta, Massimini, Deco)

Aim: This task will strongly depend on previous experiments. The goal is to create a stimulation protocol in parallel with rehabilitation to improve disconnection and dynamics post-stroke. We plan to test the hypothesis that modulation of slow wave activity (or other mechanisms of disconnection) by non-invasive stimulation improves transiently (on-line) performance. When given chronically in multiple sessions paired with rehabilitation it shall improve recovery.

Methods/Analysis: The protocol will be detailed after a thorough evaluation of preliminary results in rodents, different kinds of whole brain models, and exploratory analyses using TMS/EEG. Here we discuss some choices and trade-offs. Stroke patients. We plan to study chronic first-time stroke patients with stable deficits. Positive effects on stable symptoms would be the most convincing since it would be net of any confound of plasticity. Alternatively, stronger effects may occur in sub-acute patients when plasticity mechanisms are most active in the first three months post-stroke. This approach, however, would require a sham control group. Design. Multiple session-single subject design with multiple baseline assessments prior to treatment. Lesions and deficits: We seek a strong effect size (Cohen $d > 0.5$). We plan to use patients with right MCA lesions with significant motor deficits and visuospatial neglect. Behavioural and functional deficits will be measured as in our previous studies^{10,11}. Technique for stimulation. The protocol will be selected based on the animal work and whole brain modelling with the goal of modulating slow waves (or other behaviourally relevant mechanisms). Several options will be explored: (1) inhibitory or excitatory repetitive TMS (rTMS) to enhance or dampen slow-wave activity,²² either locally or multiple locations. There is evidence for the efficacy of theta burst stimulation to left parietal cortex in patients with visuospatial neglect.¹¹² This protocol is currently applied clinically targeting the EEG electrode P3 with no regard to the underlying pattern of disconnection or dynamic abnormalities. Accordingly, significant variability in response has been reported¹¹³. We strongly believe that at a minimum our work will permit to select the best site of stimulation based on the disconnection signal (e.g., SCD, FCD, slow waves) by abating individual variability hence giving a more stable efficacy; (2) transcranial direct or alternating current stimulation (tDCS, tACS) to inject anodal or cathodal currents vs. noise or

oscillatory responses, as in the preliminary slice data (**Fig.10**). Current commercial systems can stimulate 1-2 locations, multi-site stimulation may be coming on-line in the next 5 years. (3) EEG-guided, closed-loop stimulation on specific phases of slow waves, as in recent work and the Synergy grant Connect-2-Brain,¹¹⁴ in which Corbetta and Massimini are members of the scientific board. Loci of stimulation: based on computational and animal models. Behavioural protocols: As in rodents (task 8) we will run a reaching motor task to natural objects with the hemiparetic vs. normal arm. Arm movements will be tracked with cameras and electromyography. In these conditions patients with moderate hemiparesis cannot reach accurately or cannot shape accurately their hand. The task will be continuous to emphasize time-on task. During “reaching without stimulation”, we will first characterize the occurrence of transient slow-wave activity in relation to movement. Subsequently, we can apply “stimulation” either prior or in phase of the movement or time-locked to the slow-wave activity on some random trials. We predict that motor performance (direction, accuracy) will be worse on trials in which slow waves are stronger in motor cortex (or other disconnected areas), similarly to Andrillon et al.³⁰ Modulation of slow waves will transiently improve performance. A second protocol will record eye movements during a natural visual exploration task or goal-directed search tasks to low saliency targets. In these conditions patients with attention deficits explore the right side of the visual scene ignoring items on the left. Again, we will run a study in which we measure the on-line relationship between lateralized dynamic slow-wave transients in the parietal cortex and abnormal deviations of eye movements. Then, we can run rehabilitation protocols with stimulation.

Read-out and feedback: To stabilize behaviour, feedback may be important. Three read-out signals may be used: (1) motor/attention performance; (2) peripheral signals (EMG, trajectories, eye movements); (3) central signals (brain connectivity states; slow waves; etc.). This information can be given to the patient in a biofeedback paradigm. A more complex strategy (the one of Connect-2-Brain) is to run *closed-loop* where the feedback is based on the similarity between the patient's data and a model computed from the healthy controls.

Outcomes: We will test different strategies for intervention to improve function in stroke patients. We believe these experiments are important as they provide the state-of-the-art on circuit-based stimulation in humans. Even though an effective treatment may not be achieved, this work is fundamental to the whole idea of circuit-based stimulation. As such these experiments are critical for the field.

Dissemination

Each PI is owner and responsible for his/her own data. For each task, a series of short written agreements will be drafted to regulate collaborations, authorships, and intellectual property. All peer-reviewed scientific publications are published open access. We plan a website, a series of short video clips on the most important publications, and a short fiction documenting the project from its inception.

Risk and contingencies

Table 1 provides a list of risks and contingency plans. The major risk is that the slow wave hypothesis is not correct. However, each experiment is designed to take into consideration additional mechanisms. For example, Task 1 explores the relationship between disconnection and EEG signals not just in the low frequencies but across the spectrum (0.1 hz-100 hz). The same for iEEG (Task 2), and optical imaging/neurophysiology in rodents (Task 3). For circuit-based stimulation, any behaviourally relevant signal will do.

Task	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
1 Stroke human fMRI/EEG						
data acquisition	1/3	2/3	3/3			
analysis		SCD	FCD	SWA	Both	
publications		1	2	3		
2 Epilepsy human RFTC/SEEG						
data acquisition	1/3	2/3	3/3			
analysis		SWA	PCI	SCD	FCD	
publications		1	2	3		
3 Stroke rodent model						
data acquisition	SM	Ca++	Ca++	Ephys	Stim	
analysis			FCD	FCD	DIR	
publications		1	2	3		
4 Stroke human TMS/EEG						
data acquisition	1/3	2/3	3/3			
analysis		FCD	PCI	PCI	Both	
publications		1	2	3		
5 Slice stimulation						
data acquisition	Stim	Stim	SM	SM		
analysis		SWA	PCI	PCI		
publications		1	2	3	4	
Rodent model RFTC						
data acquisition			SM	Stim	Stim	
analysis				SWA	PCI	PCI
publications				1	2	
6 Whole brain models multi-scale						
analysis	fMRI	LFP	Ca++			
publications	BM	BM	BM	2	3	
Whole brain models stimulation						
analysis			Stim	Stim	Stim	
publications			1	2	3	
Rodent stim.+rehab						
data acquisition			Stim	Stim+ask	Stim+rehab	
8 Human stim.+rehab						
analysis				Stim+ask	Stim+rehab	
publications				1	2	3
9 Human stimulation+behavior						
analysis						
publications					1	2

Gantt chart with intermediate goals.

Task	Potential risk	Risk probability / Impact	Contingency plans / Efficacy of solution
1 Stroke human fMRI/EEG	Low recruitment stroke patients	Low / High	Recruit in other units in Padova, total #stroke/yr=600 / High
1 Stroke human fMRI/EEG	Slow-wave activity (SWA) not associated with disconnection and behavior	Medium / Medium	Find data-driven EEG signals related to disconnection and recovery / High
2 Epilepsy human RFTC/SEEG	Low recruitment thermocoagulation patients	Low / High	Recruitment is regular (20 pts per year) in Milan, but can be extended to other centers / High
3 Stroke rodent model	Disconnection, SWA, and behavior do not correlate	Medium / Medium	Identify alternative mechanisms predictive of recovery / High
4 Stroke human TMS/EEG	Difficulty in running both fMRI/EEG and EEG/TMS expts. in same subjects	Medium / Low	We will run separate expt. in different subjects / High
5 Epilepsy Rodent thermocoagulation	Adequate size of lesions and lack of detection of SWA in the perilesional region reproducing human findings	Medium/Medium	Fine adjustments with stimulation intensity and probe size. Peri-lesional physiological recordings at different distances and anatomical processing for the mapping of the affected area. Identification of the optimal conditions for long distance effects to adjust the model for clinical findings. / High
6 Whole brain models multi-scale	Model can fit data but does not predict	Medium / Medium	We will start with relative simple models assuming similar type of local dynamics. Second generation model (already under development) include more sources of heterogeneity (e.g. E/I balance, different proportion of neurotransmitters) that improve fit / High
7 Whole brain models stimulation	Stimulation prediction is not accurate	High / High	The implementation of the stimulation protocol can be refined by using more accurate simulation of the interventions in silico, i.e. individualized models of the electromagnetic stimulation by solving the explicit Maxwell equations of the stimulation set up (this solutions already exists). / Medium
8 Rodent stimulation+behavioral training	Model guided stimulation does not add to task training	Medium / High	There have been already significantly positive interventions in combination with training [e.g. botox injection in normal motor cortex]. Model will be refined using these empirical strategies. / Middle
9 Human stimulation+behavior	Model guided stimulation does not add to task training	High / high	Model refinement based on EEG readout (slow waves) and empirical strategies / Middle

Table 1. Risk and contingency plans

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Section c. Resources and time commitment (including project costs)

Cost category		Corresponding PI	2 nd PI	3 rd PI	4 th PI	Total in euro (no decimals)
PI name		Maurizio Corbetta	Gustavo Deco	Marcello Massimini	Maria V. Sanchez-Vives	
Host Institution		UNIPD	UPF	UniMi	IDIBAPS	
A. Direct personnel costs/€	PI ¹	332.603,00	432.000,00	298.488,00	358.020,00	1.421.111,00
	Senior Staff	452.611,00		300.785,00	45.000,00	798.396,00
	Post docs	819.232,00	390.000,00	280.000,00	541.008,00	2.030.240,00
	Students	424.008,00	212.000,00		202.878,00	838.886,00
	Other personnel costs		60.000,00	140.700,00	417.690,00	618.390,00
	Total personnel costs/€	2.028.454,00	1.094.000,00	1.019.973,00	1.564.596,00	5.707.023,00
B. Subcontracting Costs/€ (No indirect costs)			78.600,00			78.600,00
C. Purchase Costs/€	C.1 Travel and subsistence	100.000,00	69.000,00	45.000,00	60.000,00	274.000,00
	C.2 Equipment - including major equipment	151.000,00	88.000,00		220.000,00	469.000,00
	C.3 Other goods, works and services	Consumables incl. fieldwork and animal costs	177.600,00	9.000,00	223.500,00	360.000,00
		Publications (incl. Open Access fees) and dissemination	72.000,00	50.000,00	12.000,00	30.000,00
		Other additional direct costs	335.000,00	21.800,00	76.000,00	72.000,00
		C.3 Total other goods, works and services	584.600,00	80.800,00	311.500,00	462.000,00
	Total Purchase costs/€ (C.1 + C.2 + C.3)		835.600,00	237.800,00	356.500,00	742.000,00
D. Internally invoiced goods and services/€ (No indirect costs) ²						
E. Indirect Cost/€ e= 25% * (A + C1 + C2 + C3)		716.014,00	332.950,00	344.118,00	576.649,00	1.969.731,00
Total eligible costs/€		3.580.068,00	1.743.350,00	1.720.591,00	2.883.245,00	9.927.254,00
Requested EU contribution/€		3.580.068,00	1.743.350,00	1.720.591,00	2.883.245,00	9.927.254,00

¹ When calculating the salary, please take into account the percentage of each PI's dedicated working time to run the ERC project (i.e. minimum 30% of the working time).

² Costs for host institution invoices and invoices for other entities should be included here; e.g. access to large facilities, access to other services that are charged as unit costs.

Please indicate the duration of the project in months⁸:	72
Please indicate the % of working time each PI dedicates to the project over the period of the grant:	%
Corresponding PI name: Maurizio Corbetta	40
2nd PI name: Gustavo Deco	50
3rd PI name: Marcello Massimini	33,33
4th PI name: Maria V. Sanchez-Vives	45

BUDGET JUSTIFICATION

PI Maurizio Corbetta

A. Direct personnel costs (PI, Senior Staff, Junior Staff including post-docs and students, manager) is € 2.028.454

PI: Maurizio Corbetta will dedicate 40% of his effort to the grant. He will be responsible for the overall direction of the project, as well as supervision of the human and animal experiments at UNIPD, including the treatment proof-of-concept human protocol.

Senior staff (human experiments)

-Florinda Ferreri (h-index=32), Associate Professor of Neurology, is an international expert on the use of TMS in human neurophysiology. She will spend 10% of her time to carry out, in collaboration with Marcello Massimini, to carry out the EEG component of Task 1, TMS/EEG experiments (Task 3), and the proof-of-concept treatment trial (Task 9).

-Renzo Manara (h-index=28), Associate Professor of Neuroradiology. He will spend 10% effort to coordinate the imaging studies of the project (Task 1-3). He will be also responsible for checking the segmentation of all lesions.

-Camillo Porcaro (h-index=23), Associate Professor of Bioengineering. He will spend 10% of his time to supervise the development of analysis pipeline and statistics related to the high-density EEG/fMRI (Task 1) and TMS/EEG studies (Task 3), as well as fusion with fMRI/DTI data.

-Manfredo Atzori (h-index=16), Assistant Professor of Bioengineering. He will spend 17% of his time in developing classification algorithms to relate structural-functional disconnection, EEG, TMS/EEG feature to behavior and recovery (Task 1-3). He will be also responsible in the second part of the project to develop a model of normal performance based on behavioral, EMG, eye movements, and central EEG signals to be compared with those of stroke patients (Task 9). This comparison will be used to provide feedback to the patient during rehabilitation.

-Michele Allegra (h-index=6), Assistant Professor of Physics. He will spend 17% of his time to develop fMRI/DTI pipelines in relation to the definition of disconnection and dynamic states; fusion of fMRI/EEG data; feature selection for input into whole brain models (Task 1-3).

-Samir Suweis (h-index=19), Assistant Professor of Physics is an expert in statistical physics as applied to Neuroscience. He will spend 10% time of his time to import and develop whole brain models in Padova, and testing control theories on them.

Senior staff (animal experiments)

-Matteo Caleo (h-index=31), Professor of Physiology. He is an expert in animal physiology specifically with regards to models of stroke, epilepsy, and tumors. He will play an important role in developing the subcortical model of stroke, and in supervising the optical imaging and electrophysiological experiments in rodents (Task

3). He has designed a robotic device for mouse stroke rehabilitation that we plan to use in the proof-of-concept studies (Task 8). He will spend 12% time in the supervision and analysis of the animal experiments on stroke.

-Marco Dal Maschio (h-index=18), Associate Professor of Physiology, is a physicist with expertise in multi-modal imaging, specifically calcium imaging, and optogenetics. Dal Maschio is the coordinator of the Animal Core of the Padova Neuroscience Center. He will spend 12% of his time on running the imaging experiments, and supervision of post-docs and students.

Stefano Vassanelli (h-index=20), Associate Professor of Physiology. Expert in high-density electrophysiology and neural interfaces. He will spend 12% of his time on the electrophysiological recording and stimulation experiments in rodents including cortical and thalamic stimulation.

-Claudia Lodovichi (h-index=13), Senior Research Scientist, National Research Council, is an expert neurophysiologist. She will run and supervise the neurophysiological recordings, and supervise post-docs and students. She will spend 20% time on the project (no reimbursement)

Junior staff to-be-hired (human imaging and TMS/EEG)

-N=1 To-be-named Assistant Professor (80%). This individual will be responsible for managing and coordinating along with the PI the research activities; therefore, he will interface with both human and animal teams. In addition, he will collaborate on the integration of human and animal data through whole brain models with Gustavo Deco and Samir Suweis, the physicist in the UNIPD team. We plan to hire someone with expertise in analysis of large data sets, machine learning, and statistical physics.

-N=2 To-be named Post-doctoral Fellows (100%-time x 6 years each). 1 position will be dedicated to running MRI/DTI/EEG studies and related data analysis (Task 1). This post-doc will be supervised by Dr Allegra. The second post-doc will be dedicated to stimulation experiments with TMS/EEG. He/she will be supervised by Prof. Porcaro e Ferreri. The two positions in the second part of the project will move to the development of the stimulation protocols for rehab. They will be supervised by Drs. Allegra and Atzori.

-N=4 to-be-named PhD students (full time 3 years each; 2 for the first 3 years and 2 for the second 3 years). The pre-doctoral students will be involved in patient recruitment, behavioural testing, data processing. Each of them will be supervised by one of the post-docs.

Junior staff to-be-hired (animal experiments)

-N=3 to-be-named Post-doctoral Fellows (100% effort x 3 years each). The first post-doc will be dedicated to the development of the stroke model; the second to the imaging experiments, and the third post-doc to the neurophysiological/stimulation experiments.

-N=2 PhD students (full time 3 years each). The pre-doctoral students will be involved in animal surgeries, behavioural training, rehabilitation.

C1. Travel and subsistence

Travel - 100.000 € total in 6 years. Each year we plan 2 meetings of the whole consortium to be held on rotation in Padova, Milano, and Barcelona. We also plan 2 international conferences to which the PI, 1-2 senior and the junior members will attend. Conferences that we are targeting are Federation of European Neuroscience Societies (FENS), Society for Neuroscience (SFN), and Organization for Human Brain Mapping

C2. Equipment including major equipment

Total 151.000: 1) 71.000 €: Transcranial electrical stimulation + EEG system; 2) 80.000 €: animal rehabilitation apparatus, laser sources, objectives, amplifiers and head-stages.

C3. Other goods, works and services (total 584.600)

Consumables - total 177.600 € - Imaging (27.600): EEG caps that are prone to rupture plus supplies for EEG/TMS and MRI experiments. Each TMS/EEG cap at 128 electrodes costs 4.500 Euros. Additional material (foams, thermoplastic masks) is used during the MRI session to prevent movement. We will pay a \$500 fee/year for using the NIH toolbox for testing. Software licences (e.g. Matlab). Animal (150.000): Plasticware, glassware, reagents, neural probes and microelectrodes, fibers, optomechanical components, viral vectors.

Publications - 72.000 € in 6 years. Each open access publication is about 2.000 euro. We plan fewer publications in the first years and up to 12 publications in the last 2 years.

Other costs – Total 335.000 including audit (10.000), outreach activities (30.000) and MRI Scan (220.000 total cost for patients and healthy volunteers enrolled in the study), PNC storage+processing fee (60.000), Animal maintenance Services (15.000).

Existing resources - The project will employ numerous instruments already present at PNC-UNIPD: 3.0T Phillips Ingenia scanner; high-density EEG 128 channels integrated with TMS single/repeated stimulator with neuro-navigation; mesoscale imaging system for calcium, intrinsic, and voltage gated reporters; high density electrophysiology; and, upgraded 2P imaging for high resolution of subcellular structures. Moreover, none of

these experiments would be feasible without the clinical and technical support of the Stroke Unit and Neurology Unit of the Padova University Hospital, units from which the patients will be enrolled. Prof. Corbetta is the Chair of the Division of Neurology, and he is the clinical supervisor of these facilities. Moreover, the University Hospital of Padova has waived all costs for taking care of stroke patients enrolled in the studies.

PI Gustavo Deco

Staff - PI, senior and junior staff costs total 1.094.000 € for the PI (**time commitment 50% 6 years**), a Senior. Post doc (full-time 6 years), 2 Pre doc students (2 full-time 4-year contracts) and a research technician (part-time). Prof. Deco's team works on the whole brain modelling multi-scale and stimulation tasks. The research tech. manages data sent to the team's computational cluster, manages functioning of the cluster and workstations, and gives IT support to the researchers.

Consumables - Total 9.000 €. Small IT costs, such as 1.500€/year for disks for storing and managing experimental data.

Existing resources - Prof Deco's Computational Neuroscience group has their own computational cluster. The cluster is currently composed of 38 nodes, 14 of them new generation Intel processors, a total of 829 cores, more than 4 TB of RAM, a disk cabinet with a "snapshot" system at the user's "home" level, more than 35 TB of capacity, distributed disc support, high speed switch connections.

Equipment – Total 88.000 €. A computational cluster upgrade, 2 high performance workstations and 2 PCs are necessary. Over the course of the project, as part of the normal functioning and maintenance the cluster it is necessary to upgrade it. Also, the complexity of the simulations and the large volume of data involved in this project requires us to extend the cluster capabilities. An upgrade of 10 nodes (8.000 per node = 80.000 €) is foreseen. Again, over the course of the project, 2 new workstations and 2 new PCs would be necessary as existing IT equipment in the group reaches the end of its useful life. We upgrade and maintain PCs and workstations until the expense is prohibitive and it is more cost effective to buy new equipment. The high-performance workstations and PCs are for modelling and analysis work (2.000 per workstation + 2.000 per PC = 8.000 €).

Travel - Total 69.000 €. Team members travel to conferences. 1 EU + 1 non-EU trip per year for 4 team members. EU: *e.g.*, FENS (Forum of Neuroscience organized by Federation of European Neuroscience Societies) in Glasgow in 2020 and in Paris in 2022; ICANN (Internet Corporation for Assigned Names and Numbers) in Hamburg in 2020; the Bernstein conference (the largest annual Computational Neuroscience conference in Europe) in Berlin in 2020. Non-EU: *e.g.*, CNS (Cognitive Neuroscience Society) Boston in 2020; OHBM (Organization for Human Brain Mapping) in Montreal in 2020; SfN (Society for Neuroscience) in Washington in 2020 Chicago in 2021 and San Diego in 2022. Team travels to consortium project meetings. Travels for stays in partner labs. Team members may need to spend 1-2 months in a partner's lab for correct model development, testing and analysis.

Publications - total 50.000 €. 25 publications (high impact, Q1) are foreseen, 2.000 € per publication. Journals charge for different concepts, *e.g.*, colour figures, per page and for open access. For example, we may publish in PNAS, Cell, Trends in Neurosciences.

Other – 21.800 € total Registration fees. Conference registration for 4 team members x 2 trips per year, 1 EU + 1 non-EU conference. Registration for summer schools. Registration for special courses, for example at CRM Barcelona which provides advanced training in Mathematics. Audit (5400 €).

Subcontracting costs 78.600 €. Computational cluster maintenance. Monthly cluster maintenance service includes user management, queue management, monitoring node state, implementation of scripts, installation of programs, detection/management/resolution of incidents, documentation creation and cluster upgrade.

PI Marcello Massimini

Staff - PI, senior and junior staff: € 1.019.973

PI (Marcello Massimini MD, PhD), full professor with a commitment of 33,33%. One Associate Professor (Mario Rosanova MD, PhD), with a commitment as a co-PI of 16,66%. Two senior post-docs for supervision of analysis pipeline (Silvia Casarotto PhD in Bioengineering) and for supervision of EEG and TMS/EEG experiments (Simone Sarasso, PhD in Neuroscience) with a total commitment of 25%+33,33%. Two junior post-docs (a psychologist for the implementation of behavioural tasks and patient assessment and a bioengineer/programming, implementation of analysis pipeline) with a commitment of 50% each. Three research fellows with a background in electrophysiology (TMS/EEG experiment and scalp EEG recordings during clinical intracranial assessment) with a commitment of 65% each. One technician for project and data management with a commitment of 50%.

Consumables - Total € 223.500. 4 EEG caps/year x 5 years € 191.100 + 180 trackers for Nexstim TMS navigation system (50 patients 3 times + 30 controls = € 32.400)

Existing resources - The lab at UMIL is already equipped with the necessary hardware to run the proposed experiments. This includes two Nexstim systems for integrated TMS and MR-guided neuronavigation, two BrainAmp EEG TMS-compatible amplifiers (BrainProducts) and one high-density (256 channels) EEG system (Electrical Geodesics) for scalp EEG recordings simultaneous with intracranial recordings. We are also equipped with several workstation for data analysis.

Travel - 1 EU + 1 non-EU trip per year for 8 team members: € 45.000. EU conference: *e.g.*, FENS (Forum of Neuroscience organized by Federation of European Neuroscience Societies); European Clinical Neurophysiology, European Neurology Meeting. Non-EU conference: *e.g.*, OHBM (Organization for Human Brain Mapping), SfN (Society for Neuroscience Meeting), Team travel to consortium project meetings. Travel for stays in partner labs.

Publications - open access 3/4 papers on peer-reviewed journals (€ 12.000)

Other good and services – Total 76.000 € including audit (€ 17.500) 3 workshops' organization (€ 22.500) and fees for conference, workshop and school (30.000 €) external IT service, data storage (€ 6.000).

PI Maria V. Sanchez-Vives

Total personnel costs: 1,564,596

Senior Staff

Dr. Sanchez-Vives is an ICREA (**Institució Catalana de Recerca i Estudis Avançats**) Researcher. She will have a dedication of at least 45% of her time. ICREA is a foundation supported by the Catalan Government and guided by a Board of Trustees which aims to recruit top scientists for the Catalan R&D system: scientists capable of leading new research groups, strengthening existing groups, and setting up new lines of research. IDIBAPS has a formal relationship with ICREA, who will provide third-party in-kind resources (Prof. Sanchez-Vives salary) to IDIBAPS. Although the salary costs of Prof Sanchez-Vives are paid by ICREA, she is assigned to physically work at IDIBAPS and considered a member of IDIBAPS. Dr Sanchez-Vives will be the coordinator of IDIBAPS research in NEMESIS.

Dr. Anna Planas is Head of the Department of Brain Ischemia and Neurodegeneration at the Institute for Biomedical Research (IIBB-CSIC) and Coordinator of the Neuroscience Area at the Institut d'Investigacions Biomèdiques August Pi i Sunyer (IDIBAPS). She has an h-index of 63, is an expert on animal models of stroke and will participate in the project.

Dr. Pedro Roldán, Dr. Jordi Rumiá, Neurosurgeons at Hospital Clínic de Barcelona (HCB), **Dr. Mar Carreño** Head of Epilepsy and **Dr. Estefania Conde** from the same unit will also be involved in NEMESIS. Hospital Clínic de Barcelona (HCB) is an affiliated entity of IDIBAPS. HCB contribution on the task 5 (collaboration regarding research in human tissue) has a personnel cost of 45,000 EUR and they have been included within IDIBAPS budget.

Junior Staff to be hired

- Postdoctoral Researcher Diana Casas has experience in electrophysiology in *in vitro* preparations and is a senior investigator at Dr. Sanchez-Vives' laboratory (72 PM - 270,504 €).

- A postdoctoral position will be open. The research team requires rather high specialization on electrophysiology in vivo (anesthetized and head-fixed preparations), as well as experience on imaging techniques and ideally on neuromodulation (72 PM - 270,504 €).

- Data analyst: Arnau Manasanch is an experienced engineer and data analyzer of brain recordings at Dr Sanchez-Vives' laboratory. He would devote 100% of his time. (72 PM - 270,504 €).

- A predoctoral position will be open during the 6 years of the project, which will probably allow 2 different predocs to obtain their PhD degrees within the duration of NEMESIS, given that standard PhD duration is now in Spain 3-3.5 years (72 PM - 202,878 €).

- A lab/project manager will devote 15% of his/her time to all the tasks related to NEMESIS, acquisition and maintenance of equipment, administration, contracts, trip organization, interfacing with IDIBAPS, accounting, technical/financial reporting, coordination of personnel, etc. (10.8 PM - 47,736 €).

- A lab technician with 50% dedication will support on laboratory procedures, in vivo and in vitro preparations and supervision of animals, animal perfusion, anatomical processing and reconstructions, laboratory maintenance, coordination with animal facilities (36 PM - 99,450 €).

Consumables – Total 360,000 € - The laboratory consumables include animal costs and maintenance, laboratory materials (including ad hoc-designed electrode arrays, microtransistor arrays, drugs, anesthetics, implants, electronics).

Equipment – Total 220,000 € - The funds required for equipment include informatic material (updates of some computers, licenses, computers for new hirings) and completion or update of electrophysiological set ups (light sources, peristaltic pump, temperature probe, a generator system for radiofrequency thermo-lesions, new CCD camera, antivibratory table, micromirror device and high power LED for patterned illumination (photopharmacology), multichannel system for wireless chronic recordings, ad hoc-designed new tDCS systems for in vivo, needs that emerge and maintenance). Notice that our planned experiments require at least 2 *in vitro* set ups (we often use 3 simultaneously to optimize the use of brain slices), 1 for acute and 1 for chronic/head fixed.

Existing resources - IDIBAPS is provided with all the necessary laboratories, equipment and scientific support required for its researchers. IDIBAPS offers several last-generation scientific and technological services including Confocal and Electron Microscopy, Genomics, Ultrasequencing, Proteomics (including ultrasequencing capacity, Cytometry, Bioinformatics, Mass spectrometry, Microscopy and Life Cell imaging Unit, Image Analysis Unit (including PET or 7T fMRI for animal systems), Cytometry and Cell separation Unit, Clinical Trials Unit and Histology among other advanced facilities. A complete list of the research support services (Platforms) available to IDIBAPS researchers can be found at http://www.idibaps.org/en_index.html. The institute is also equipped with molecular biology laboratories (studies of DNA and RNA, siRNA design), biology of lipids and immunohistology, as well as dedicated facilities for the preparation, culturing and maintenance of primary cells and the realization of studies on cellular cultures of human as well as of animal origin. Specialized scientists are responsible for these laboratories and are in charge of offering scientific and technical support to the different research domains of IDIBAPS. The animal facility of the University of Barcelona is also available for the personnel working at IDIBAPS. This facility is supervised by specialized veterinary doctors, which is set up for developing experimental models in pigs, rats and rabbits, as well as for developing and breeding genetically modified animals.

The laboratory of Dr. Sanchez-Vives has several *in vivo* and *in vitro* set ups that allow different configurations of electrophysiological recordings. However, to carry out the proposed experiments, some updates and novel equipment is needed and has been mentioned in Equipment.

Travel – 60,000 € have been estimated for annual trips to international conferences and to meetings of the consortia.

Publications – Total 30,000 € - We have estimated an average of 5,000 € / year given the current costs of open access publications and that some will be shared with partners.

Other goods and services – – Total 72,000 € that includes audit costs (28,000 €) and animal facility services (44,000 €).